

Name: Laiba Mudassar

Roll No: SP26-BCS-021

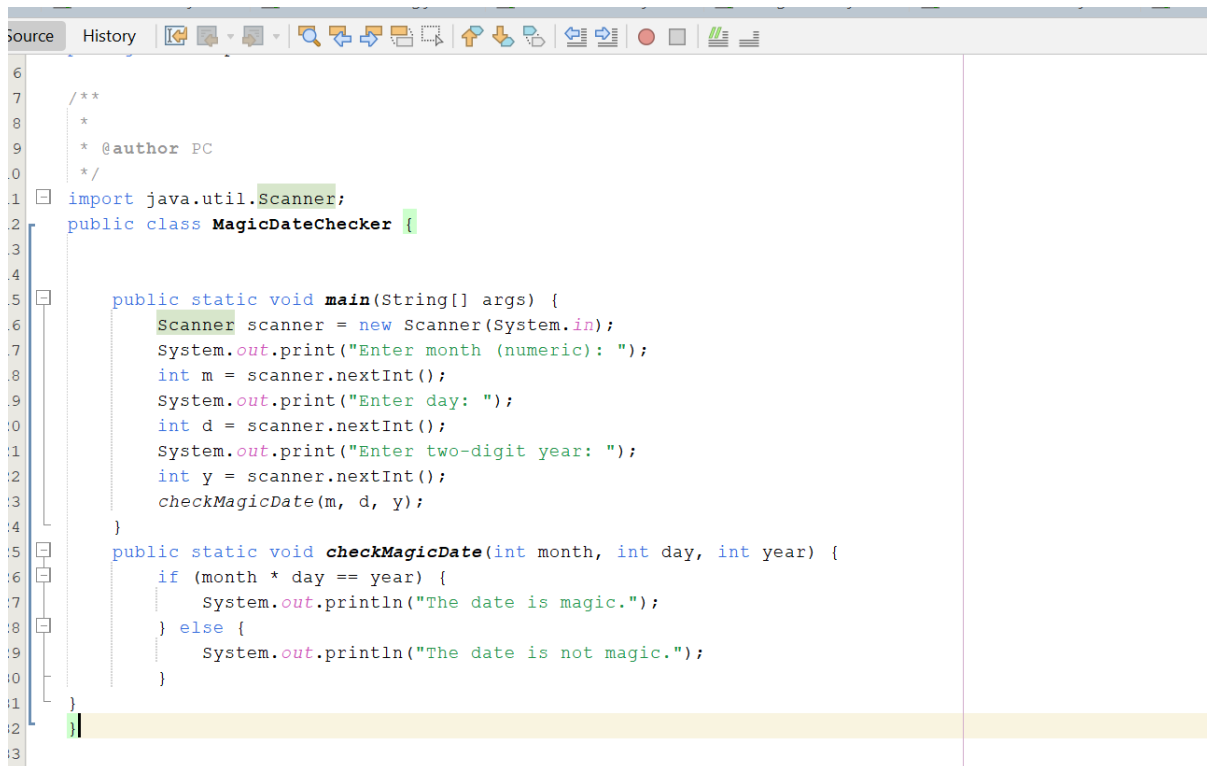
Course: Programming Fundamentals

Assignment: Assignment # 03 Theory

Submitted To: Sir Mudassar

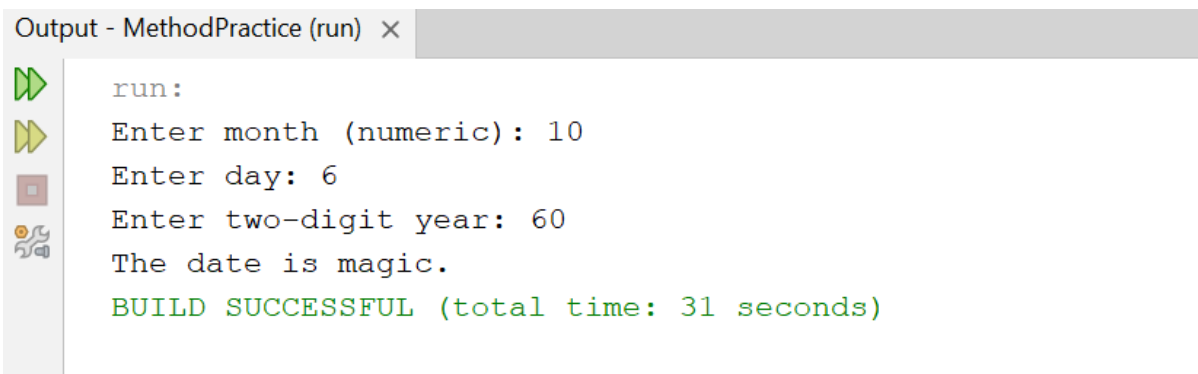
COMSATS University Islamabad

Question no 1



```
6
7  /**
8   *
9   * @author PC
10  */
11  import java.util.Scanner;
12  public class MagicDateChecker {
13
14      public static void main(String[] args) {
15          Scanner scanner = new Scanner(System.in);
16          System.out.print("Enter month (numeric): ");
17          int m = scanner.nextInt();
18          System.out.print("Enter day: ");
19          int d = scanner.nextInt();
20          System.out.print("Enter two-digit year: ");
21          int y = scanner.nextInt();
22          checkMagicDate(m, d, y);
23      }
24      public static void checkMagicDate(int month, int day, int year) {
25          if (month * day == year) {
26              System.out.println("The date is magic.");
27          } else {
28              System.out.println("The date is not magic.");
29          }
30      }
31  }
```

Output



```
Output - MethodPractice (run) X
run:
Enter month (numeric): 10
Enter day: 6
Enter two-digit year: 60
The date is magic.
BUILD SUCCESSFUL (total time: 31 seconds)
```

Question No 2

```
...va  MaxOverloading.java X  ReverseMethod.java X  StringMethod.java X  RecursionFictorial.java X  fibnoccSeries.java X  MagicDateC
Source  History  [Icons]
9      * @author PC
10     */
11     import java.util.Scanner;
12     public class FictorialBoth {
13
14     public static void main(String[] args) {
15         Scanner scanner = new Scanner(System.in);
16         System.out.print("Enter an integer: ");
17         int num = scanner.nextInt();
18         System.out.println("Loop Result: " + factorialLoop(num));
19         System.out.println("Recursive Result: " + factorialRecursive(num));
20     }
21     public static int factorialLoop(int n) {
22         int fact = 1;
23         for (int i = 1; i <= n; i++) {
24             fact = fact * i;
25         }
26         return fact;
27     }
28
29     public static int factorialRecursive(int n) {
30         if (n == 0 || n == 1) {
31             return 1;
32         }
33         return n * factorialRecursive(n - 1);
34     }
35 }
```

Output - MethodPractice (run) X

Output - MethodPractice (run)

Enter an integer: 4

Loop Result: 24

Recursive Result: 24

BUILD SUCCESSFUL (total time: 12 seconds)

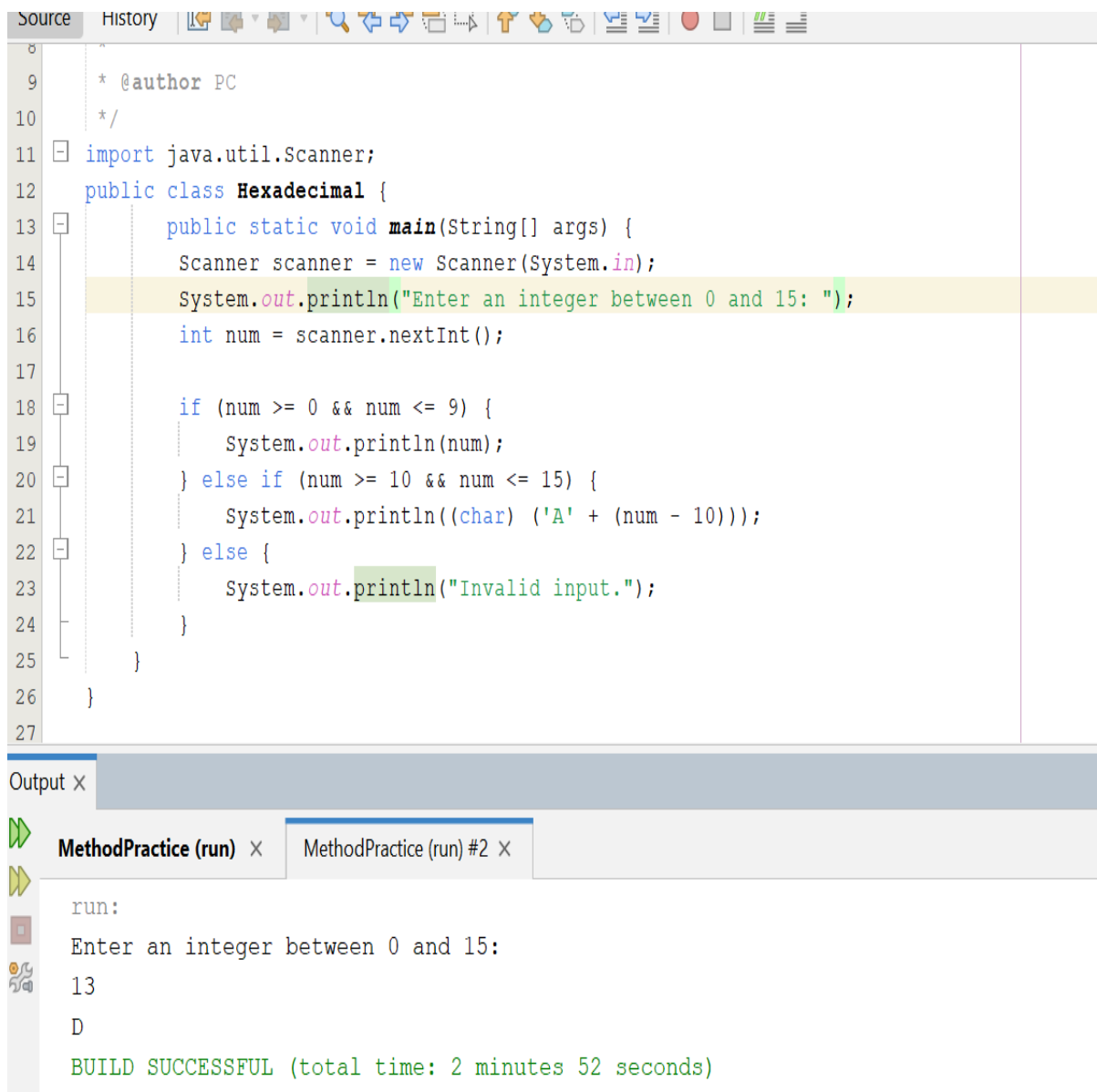
Question No 3

```
9      ^ @author PC
10     */
11     import java.util.Scanner;
12     public class AsciiCode {
13     public static void main(String[] args) {
14         Scanner scanner = new Scanner(System.in);
15
16         System.out.print("Enter an ASCII code (0-127): ");
17         int asciiCode = scanner.nextInt();
18         System.out.println((char) asciiCode);
19
20         System.out.print("Enter a character: ");
21         char ch = scanner.next().charAt(0);
22         System.out.println((int) ch);
23     }
24 }
25
26
```

Output - MethodPractice (run) ×

```
run:
Enter an ASCII code (0-127): 64
@
Enter a character: /
47
BUILD SUCCESSFUL (total time: 14 seconds)
```

Question No 4



The screenshot shows an IDE with a Java source file. The code defines a class `Hexadecimal` with a `main` method. The `main` method uses a `Scanner` to read an integer from the user. It then checks if the integer is between 0 and 9, 10 and 15, or invalid. If valid, it prints the integer or its hexadecimal equivalent (A-F). If invalid, it prints "Invalid input.".

```
8
9  * @author PC
10 */
11 import java.util.Scanner;
12 public class Hexadecimal {
13     public static void main(String[] args) {
14         Scanner scanner = new Scanner(System.in);
15         System.out.println("Enter an integer between 0 and 15: ");
16         int num = scanner.nextInt();
17
18         if (num >= 0 && num <= 9) {
19             System.out.println(num);
20         } else if (num >= 10 && num <= 15) {
21             System.out.println((char) ('A' + (num - 10)));
22         } else {
23             System.out.println("Invalid input.");
24         }
25     }
26 }
27
```

The output window shows the following text:

```
run:
Enter an integer between 0 and 15:
13
D
BUILD SUCCESSFUL (total time: 2 minutes 52 seconds)
```

Question No 5

```
package methodpractice;

/**
 *
 * @author PC
 */
public class RandomUpperCsae {
    public static void main(String[] args) {
        int randomAscii = 65 + (int)(Math.random() * 26);
        System.out.println((char) randomAscii);
    }
}
```

put ×

MethodPractice (run) ×

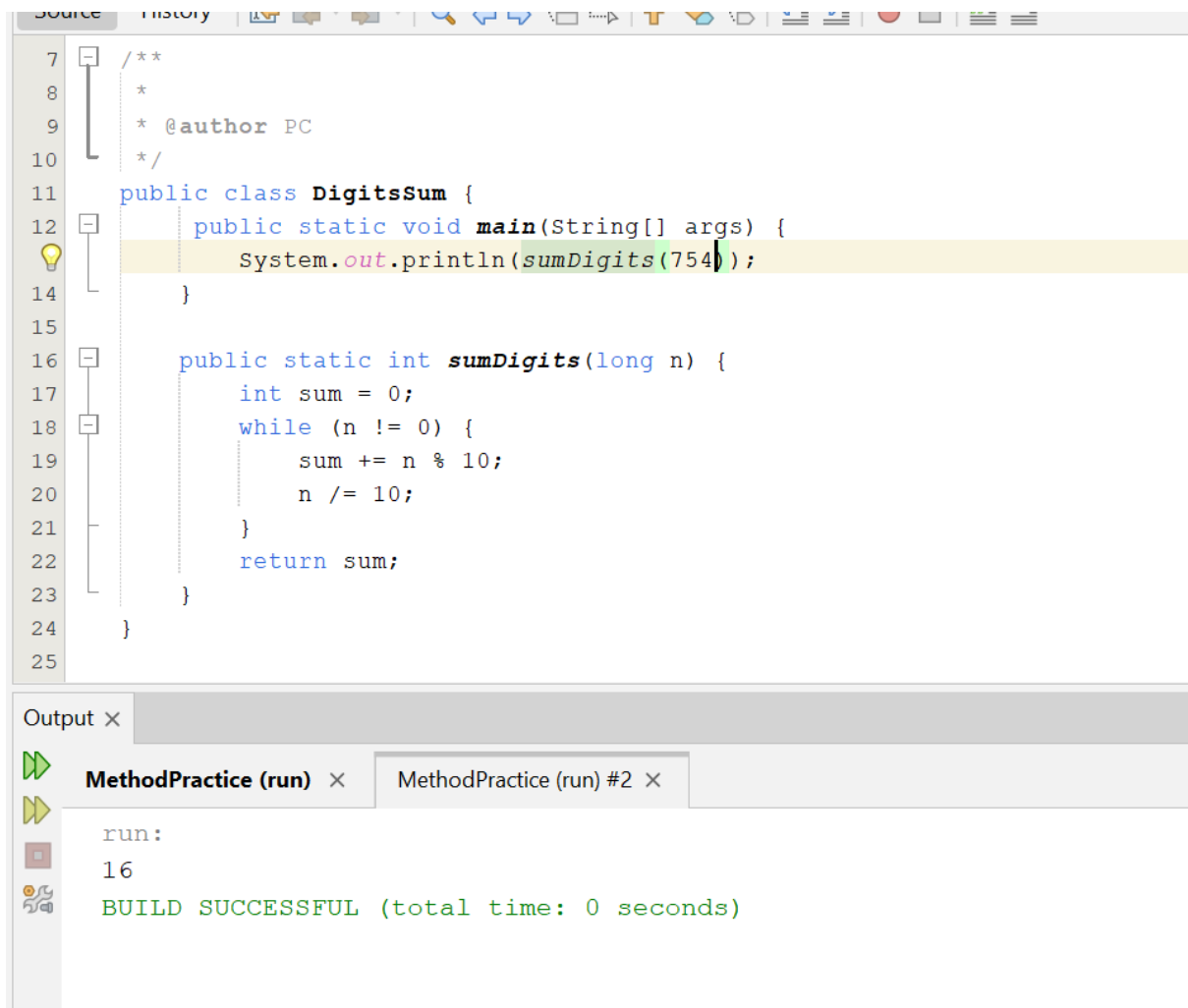
MethodPractice (run) #2 ×

run:

○

BUILD SUCCESSFUL (total time: 0 seconds)

Question No 6



The screenshot displays an IDE with a Java source file and its output. The source code defines a class `Digitssum` with a `main` method and a static `sumDigits` method. The `main` method calls `sumDigits(754)`, and the output shows the result `16`.

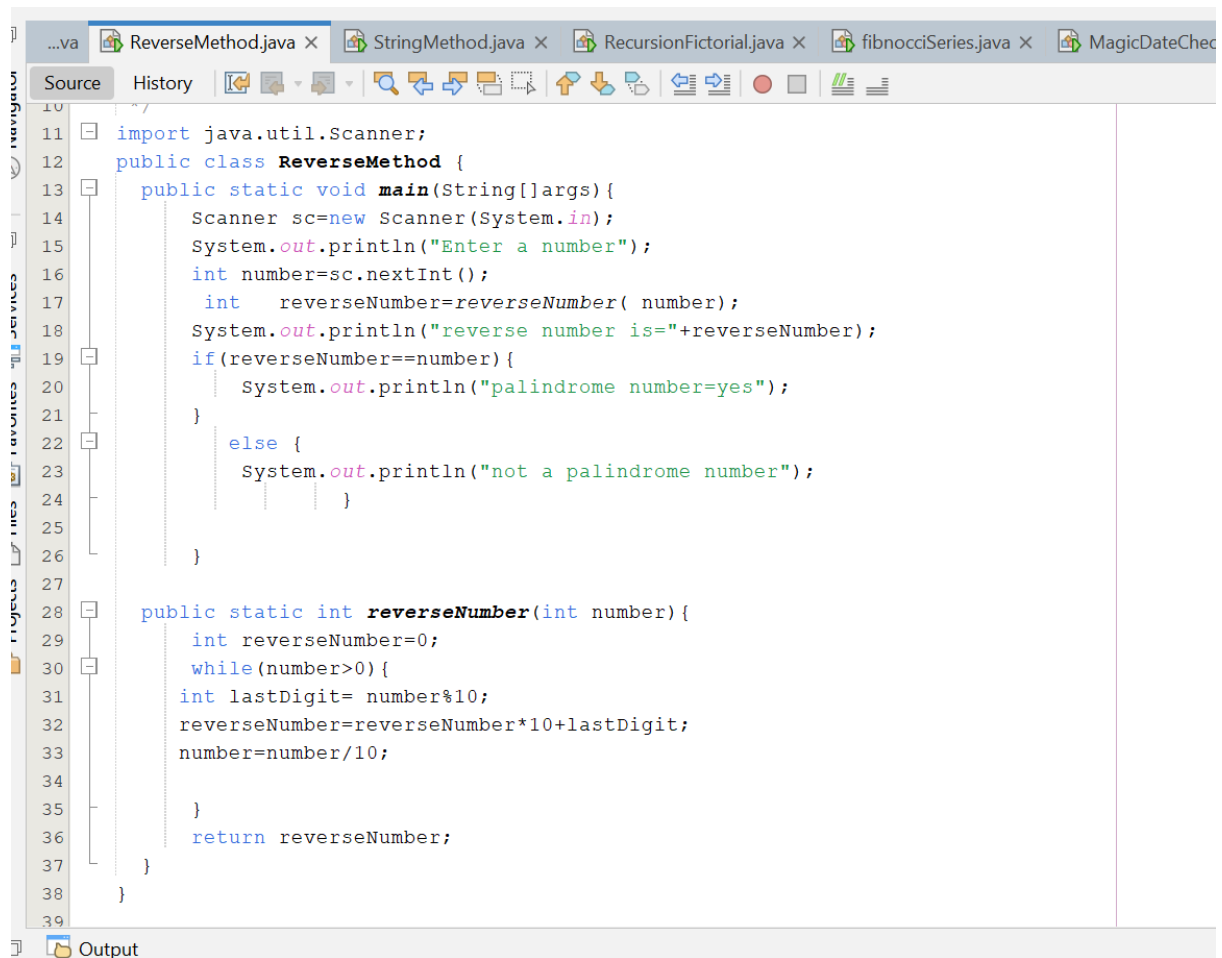
```
7  /**
8   *
9   * @author PC
10  */
11  public class Digitssum {
12      public static void main(String[] args) {
13          System.out.println(sumDigits(754));
14      }
15
16      public static int sumDigits(long n) {
17          int sum = 0;
18          while (n != 0) {
19              sum += n % 10;
20              n /= 10;
21          }
22          return sum;
23      }
24  }
25
```

Output ×

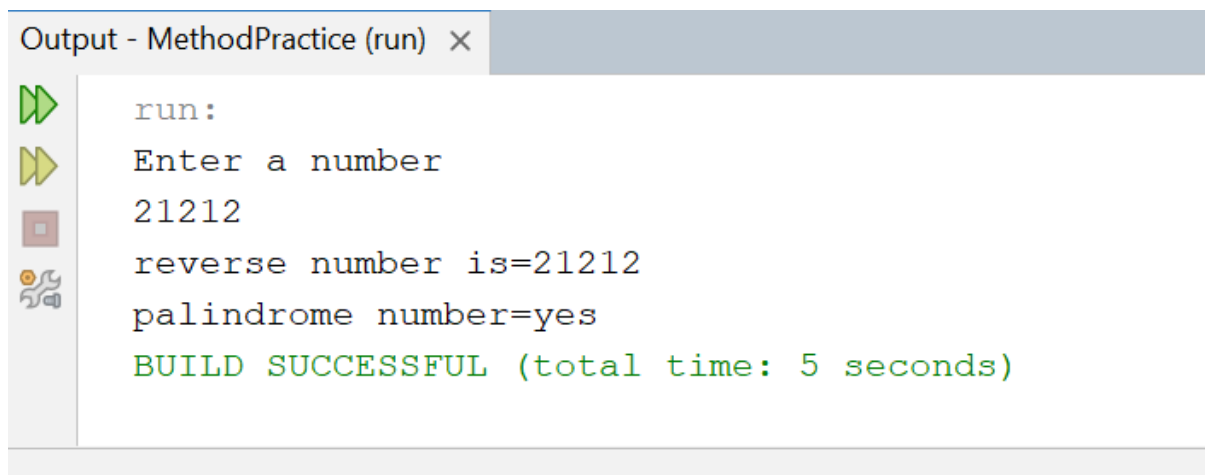
MethodPractice (run) × MethodPractice (run) #2 ×

run:
16
BUILD SUCCESSFUL (total time: 0 seconds)

Question No7

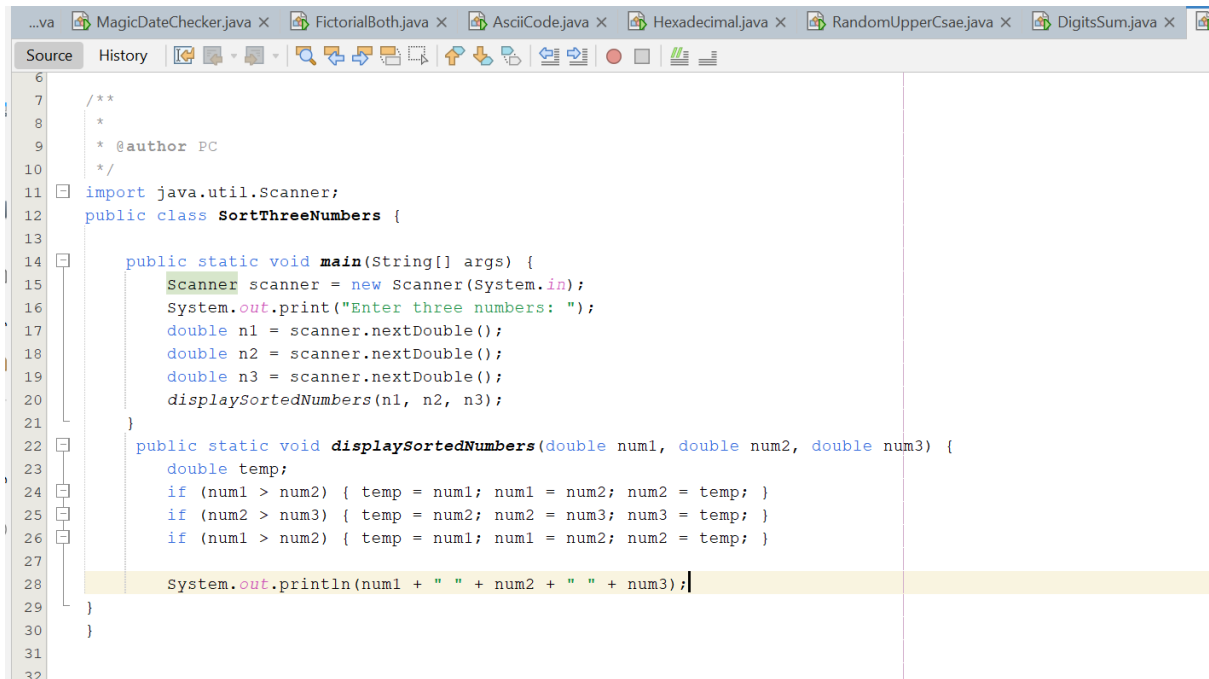


```
10
11 import java.util.Scanner;
12 public class ReverseMethod {
13     public static void main(String[] args) {
14         Scanner sc=new Scanner(System.in);
15         System.out.println("Enter a number");
16         int number=sc.nextInt();
17         int reverseNumber=reverseNumber( number);
18         System.out.println("reverse number is="+reverseNumber);
19         if(reverseNumber==number) {
20             System.out.println("palindrome number=yes");
21         }
22         else {
23             System.out.println("not a palindrome number");
24         }
25     }
26 }
27
28 public static int reverseNumber(int number) {
29     int reverseNumber=0;
30     while(number>0){
31         int lastDigit= number%10;
32         reverseNumber=reverseNumber*10+lastDigit;
33         number=number/10;
34     }
35     return reverseNumber;
36 }
37 }
38
39
```

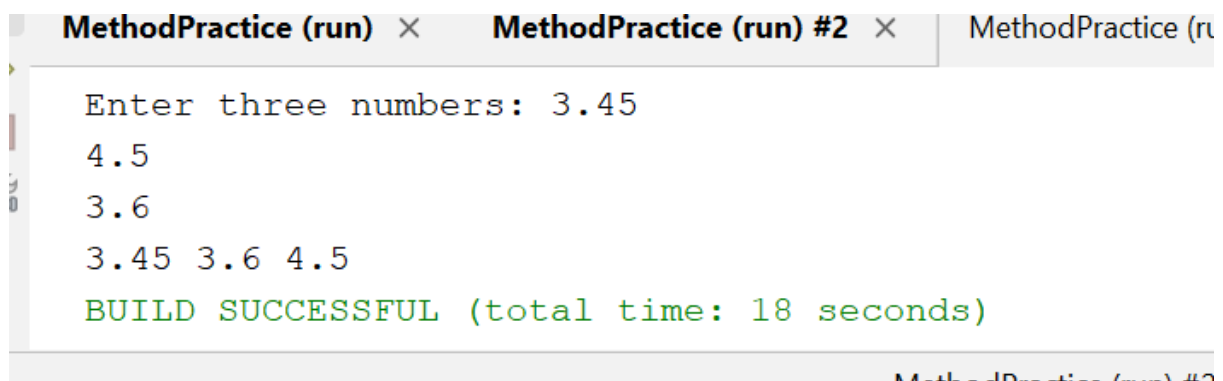


```
Output - MethodPractice (run) x
run:
Enter a number
21212
reverse number is=21212
palindrome number=yes
BUILD SUCCESSFUL (total time: 5 seconds)
```

Question No 8

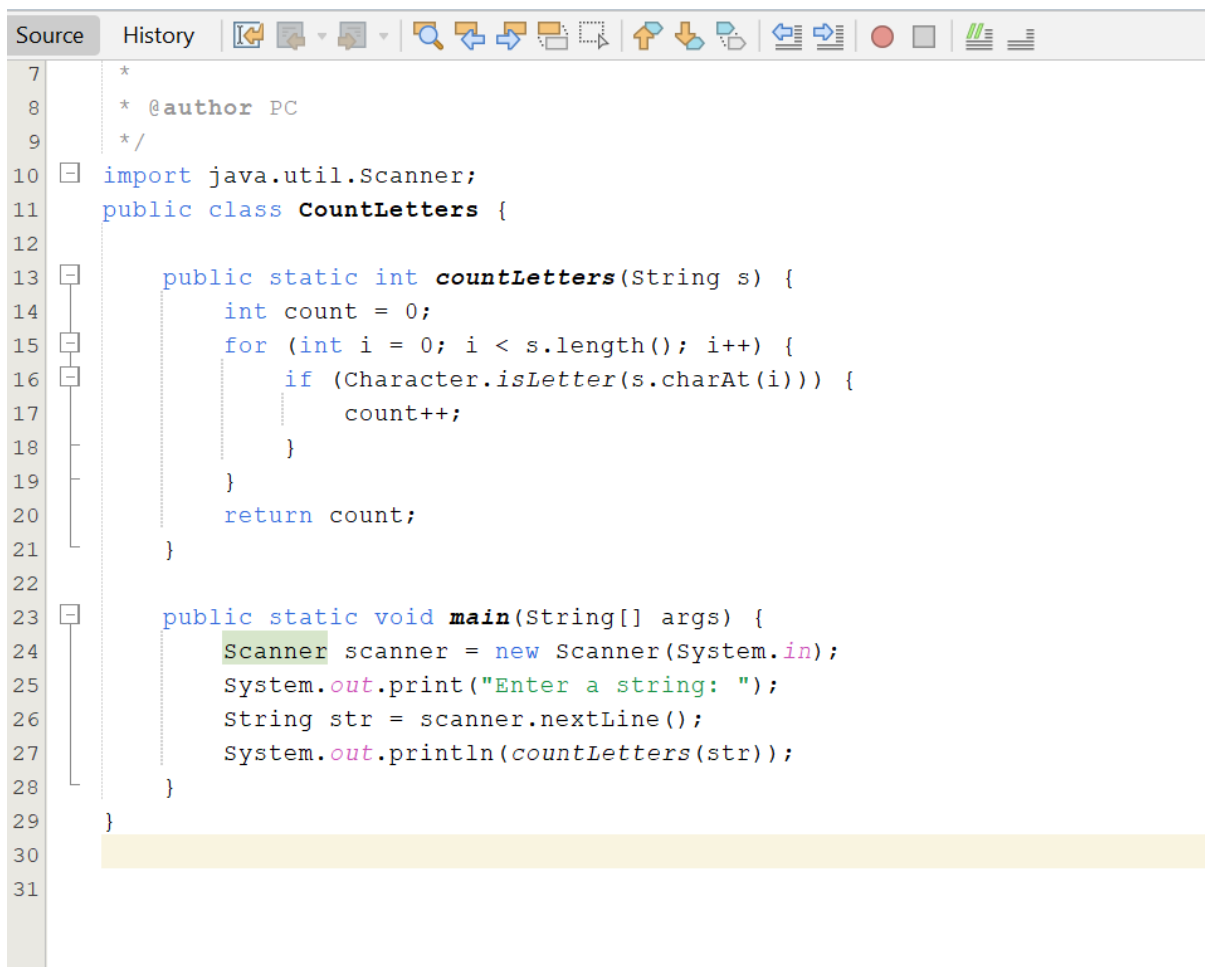


```
6
7  /**
8   *
9   * @author PC
10  */
11  import java.util.Scanner;
12  public class SortThreeNumbers {
13
14      public static void main(String[] args) {
15          Scanner scanner = new Scanner(System.in);
16          System.out.print("Enter three numbers: ");
17          double n1 = scanner.nextDouble();
18          double n2 = scanner.nextDouble();
19          double n3 = scanner.nextDouble();
20          displaySortedNumbers(n1, n2, n3);
21      }
22
23      public static void displaySortedNumbers(double num1, double num2, double num3) {
24          double temp;
25          if (num1 > num2) { temp = num1; num1 = num2; num2 = temp; }
26          if (num2 > num3) { temp = num2; num2 = num3; num3 = temp; }
27          if (num1 > num2) { temp = num1; num1 = num2; num2 = temp; }
28          System.out.println(num1 + " " + num2 + " " + num3);
29      }
30  }
31
32
```

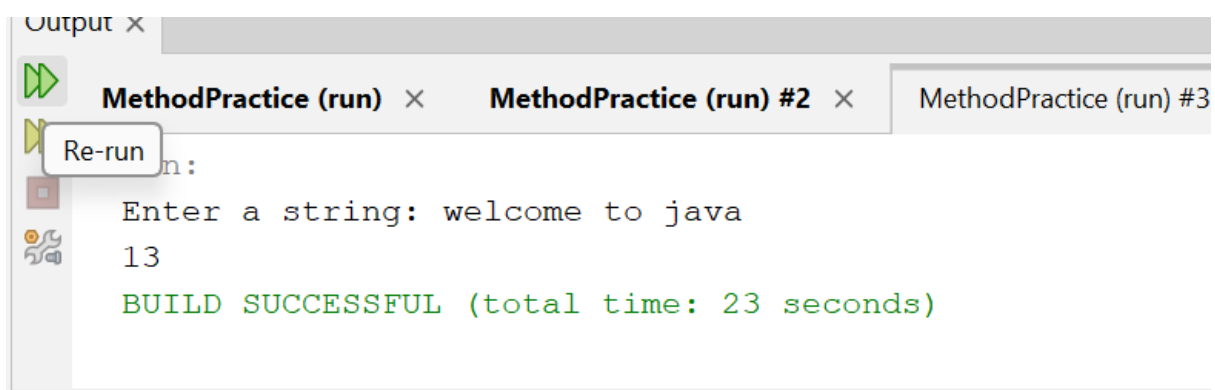


```
MethodPractice (run) × MethodPractice (run) #2 × MethodPractice (run)
Enter three numbers: 3.45
4.5
3.6
3.45 3.6 4.5
BUILD SUCCESSFUL (total time: 18 seconds)
MethodPractice (run) #2
```

Question No9



```
7      *
8      * @author PC
9      */
10     import java.util.Scanner;
11     public class CountLetters {
12
13         public static int countLetters(String s) {
14             int count = 0;
15             for (int i = 0; i < s.length(); i++) {
16                 if (Character.isLetter(s.charAt(i))) {
17                     count++;
18                 }
19             }
20             return count;
21         }
22
23         public static void main(String[] args) {
24             Scanner scanner = new Scanner(System.in);
25             System.out.print("Enter a string: ");
26             String str = scanner.nextLine();
27             System.out.println(countLetters(str));
28         }
29     }
30
31
```



Output x

MethodPractice (run) x MethodPractice (run) #2 x MethodPractice (run) #3

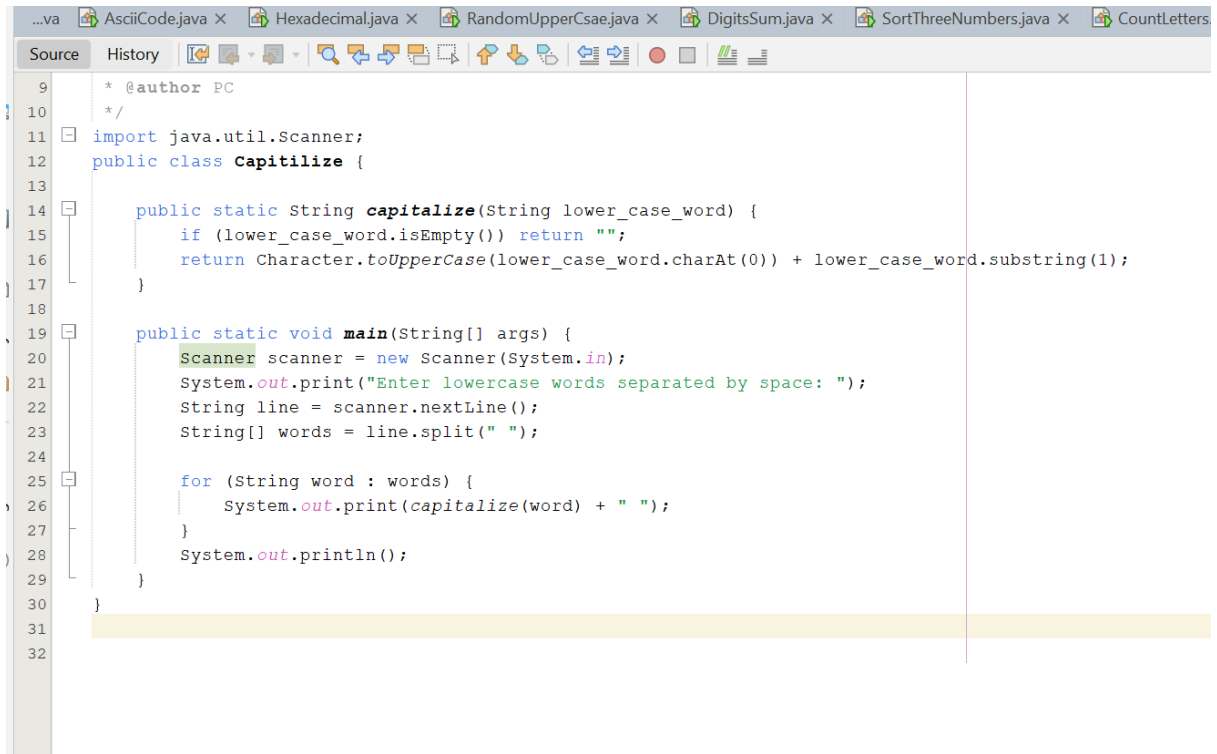
Re-run

Enter a string: welcome to java

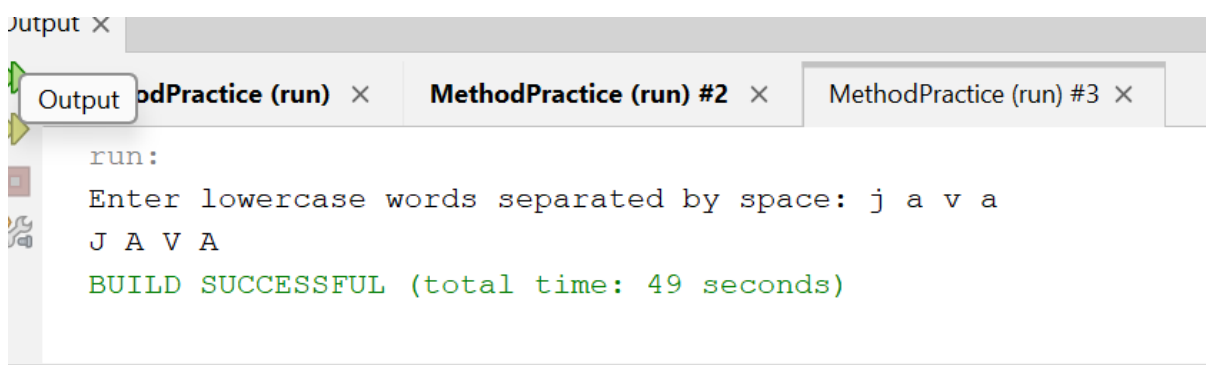
13

BUILD SUCCESSFUL (total time: 23 seconds)

Question No 10



```
9      * @author PC
10     */
11     import java.util.Scanner;
12     public class Capitalize {
13
14         public static String capitalize(String lower_case_word) {
15             if (lower_case_word.isEmpty()) return "";
16             return Character.toUpperCase(lower_case_word.charAt(0)) + lower_case_word.substring(1);
17         }
18
19         public static void main(String[] args) {
20             Scanner scanner = new Scanner(System.in);
21             System.out.print("Enter lowercase words separated by space: ");
22             String line = scanner.nextLine();
23             String[] words = line.split(" ");
24
25             for (String word : words) {
26                 System.out.print(capitalize(word) + " ");
27             }
28             System.out.println();
29         }
30     }
31
32
```



```
run:
Enter lowercase words separated by space: j a v a
J A V A
BUILD SUCCESSFUL (total time: 49 seconds)
```

Question No 11

```
1  * @author PC
2  */
3  public class VowelsCount {
4      public static int countVowels(String s) {
5          int count = 0;
6          String lowerS = s.toLowerCase();
7          for (int i = 0; i < lowerS.length(); i++) {
8              char c = lowerS.charAt(i);
9              if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') {
10                  count++;
11              }
12          }
13          return count;
14      }
15
16      public static void main(String[] args) {
17          System.out.println(countVowels("Hello World"));
18      }
19  }
```

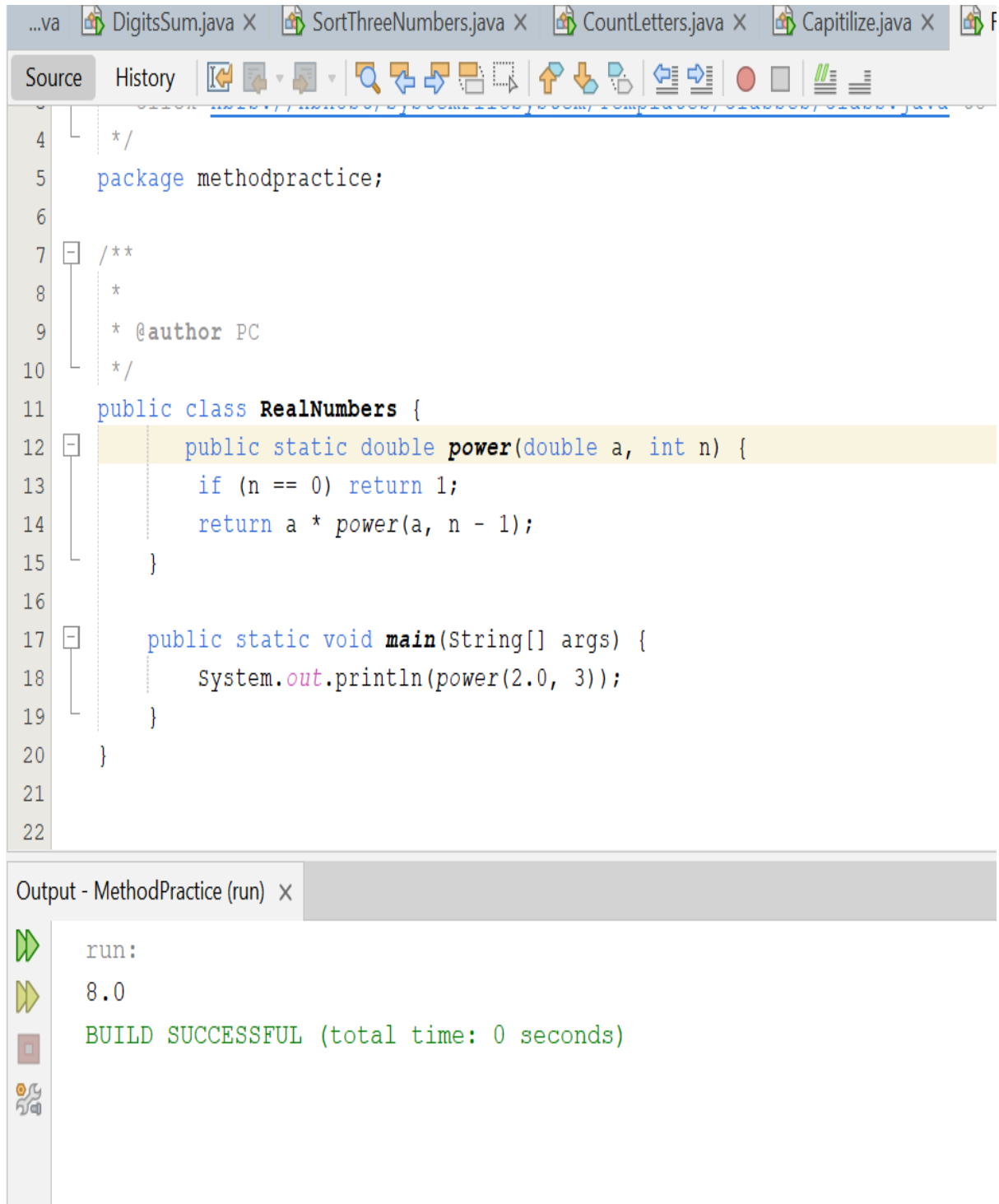
tput - MethodPractice (run) ×

run:

3

BUILD SUCCESSFUL (total time: 0 seconds)

Question No 12



The screenshot shows an IDE with several open files: DigitsSum.java, SortThreeNumbers.java, CountLetters.java, and Capitalize.java. The active file is a Java program named RealNumbers.java. The code defines a package methodpractice and a class RealNumbers. The class contains a recursive static method power that calculates the power of a double 'a' to the integer 'n'. The main method calls power(2.0, 3) and prints the result. The output window shows the result 8.0 and a successful build.

```
4  */
5  package methodpractice;
6
7  /**
8   *
9   * @author PC
10  */
11  public class RealNumbers {
12      public static double power(double a, int n) {
13          if (n == 0) return 1;
14          return a * power(a, n - 1);
15      }
16
17      public static void main(String[] args) {
18          System.out.println(power(2.0, 3));
19      }
20  }
21
22
```

Output - MethodPractice (run) x

run:
8.0
BUILD SUCCESSFUL (total time: 0 seconds)

Question No 13

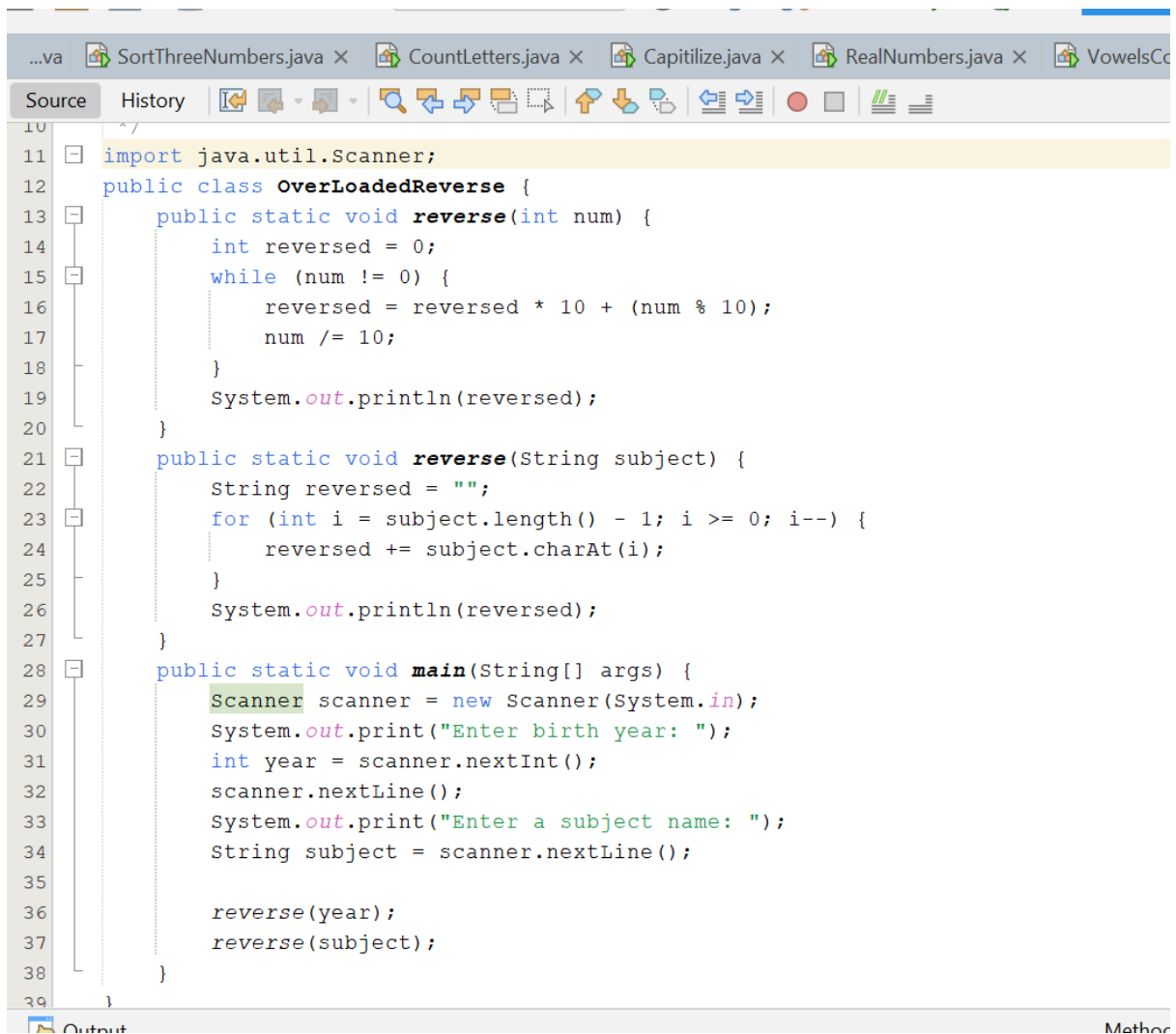
```
5 package methodpractice;
6
7 /**
8  *
9  * @author PC
10  */
11 import java.util.Scanner;
12 public class AreaOfPentagon {
13     public static double area(double side) {
14         return (5 * side * side) / (4 * Math.tan(Math.PI / 5));
15     }
16     public static void main(String[] args) {
17         Scanner scanner = new Scanner(System.in);
18         System.out.println("Enter the side: ");
19         double s = scanner.nextDouble();
20         System.out.println("The area of the pentagon is " + area(s));
21     }
22 }
23
24
```

Output ×

MethodPractice (run) × MethodPractice (run) #2 ×

Enter the side:
4.5
The area of the pentagon is 34.83966736192658
BUILD SUCCESSFUL (total time: 7 seconds)

Question No 14



```
10  */
11  import java.util.Scanner;
12  public class OverLoadedReverse {
13      public static void reverse(int num) {
14          int reversed = 0;
15          while (num != 0) {
16              reversed = reversed * 10 + (num % 10);
17              num /= 10;
18          }
19          System.out.println(reversed);
20      }
21      public static void reverse(String subject) {
22          String reversed = "";
23          for (int i = subject.length() - 1; i >= 0; i--) {
24              reversed += subject.charAt(i);
25          }
26          System.out.println(reversed);
27      }
28      public static void main(String[] args) {
29          Scanner scanner = new Scanner(System.in);
30          System.out.print("Enter birth year: ");
31          int year = scanner.nextInt();
32          scanner.nextLine();
33          System.out.print("Enter a subject name: ");
34          String subject = scanner.nextLine();
35
36          reverse(year);
37          reverse(subject);
38      }
39  }
```

tput x

MethodPractice (run) x

MethodPractice (run) #2 x

Enter birth year: 2006

Enter a subject name: programming

6002

gnimmargarp

BUILD SUCCESSFUL (total time: 27 seconds)

Mo

Question No 15

```
...va CountLetters.java × Capitalize.java × RealNumbers.java × VowelsCount.java × AreaOfPentagon.java × OverLoadedReverse
Source History
5 package methodpractice;
6
7 /**
8  *
9  * @author PC
10 */
11 import java.util.Scanner;
12 public class ShapeArea {
13     public static double calculateArea(double radius) {
14         return Math.PI * radius * radius;
15     }
16
17     public static double calculateArea(double length, double width) {
18         return length * width;
19     }
20
21     public static double calculateArea(double base, double height, String shape) {
22         return 0.5 * base * height;
23     }
24
25     public static void main(String[] args) {
26         Scanner scanner = new Scanner(System.in);
27         System.out.println("Choose shape: 1. Circle 2. Rectangle 3. Triangle");
28         int choice = scanner.nextInt();
29
30         switch (choice) {
31             case 1:
32                 System.out.print("Enter radius: ");
33                 double r = scanner.nextDouble();
```

```
Scanner scanner = new Scanner(System.in);
System.out.println("Choose shape: 1. Circle 2. Rectangle 3. Triangle");
int choice = scanner.nextInt();

switch (choice) {
    case 1:
        System.out.print("Enter radius: ");
        double r = scanner.nextDouble();
        System.out.println(calculateArea(r));
        break;
    case 2:
        System.out.print("Enter length and width: ");
        double l = scanner.nextDouble();
        double w = scanner.nextDouble();
        System.out.println(calculateArea(l, w));
        break;
    case 3:
        System.out.print("Enter base and height: ");
        double b = scanner.nextDouble();
        double h = scanner.nextDouble();
        System.out.println(calculateArea(b, h, "Triangle"));
        break;
    default:
        System.out.println("Invalid choice");
}
```

```
MethodPractice (run) × MethodPractice (run) #2 ×
Choose shape: 1. Circle 2. Rectangle 3. Triangle
1
Enter radius: 4.5
63.61725123519331
BUILD SUCCESSFUL (total time: 10 seconds)
```

Question no 16

```
* @author PC
*/
import java.util.Scanner;
public class LargestArray {
    public static int largest(int[] arr, int index) {
        if (index == arr.length - 1) return arr[index];
        int maxInRest = largest(arr, index + 1);
        return (arr[index] > maxInRest) ? arr[index] : maxInRest;
    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int[] numbers = new int[8];
        System.out.print("Enter 8 integers: ");
        for (int i = 0; i < 8; i++) {
            numbers[i] = scanner.nextInt();
        }
        System.out.println("Largest: " + largest(numbers, 0));
    }
}
```

```
MethodPractice (run) × MethodPractice (run) #2 × MethodPractice (run) #3 ×
45
34
56
89
Largest: 89
```

Question No 17

```

    * @author PC
    */
    public class ReversiveSeries {

        public static double m(int i) {
            if (i == 1) return 1.0 / 3.0;
            return (double) i / (2 * i + 1) + m(i - 1);
        }

        public static void main(String[] args) {
            for (int i = 1; i <= 10; i++) {
                System.out.println("m(" + i + ") = " + m(i));
            }
        }
    }

```

MethodPractice (run) × MethodPractice (run) #2 × MethodPractice (run)

```

run:
m(1) = 0.3333333333333333
m(2) = 0.7333333333333334
m(3) = 1.161904761904762
m(4) = 1.6063492063492064
m(5) = 2.060894660894661
m(6) = 2.5224331224331227
m(7) = 2.9890997890997895
m(8) = 3.459688024393907
m(9) = 3.933372234920223
m(10) = 4.409562711110699
BUILD SUCCESSFUL (total time: 0 seconds)

```

Question No 18

```
*/
import java.util.Scanner;
public class fibnoccSeries {
    public static void main(String[]args){
        Scanner sc=new Scanner(System.in);
        System.out.println("enter a index for a fibnocci series");
        int index=sc.nextInt();
        System.out.println("the fibonacci number at index"+index+"is"+fib(index));
    }
    public static long fib(long index){
        if(index==0)
            return 0;
        else if(index==1)
            return 1;
        else
            return fib(index-1)+fib(index-2);
    }
}
```

ut X

MethodPractice (run) X

MethodPractice (run) #2 X

MethodPractice (run) #3 X

run:

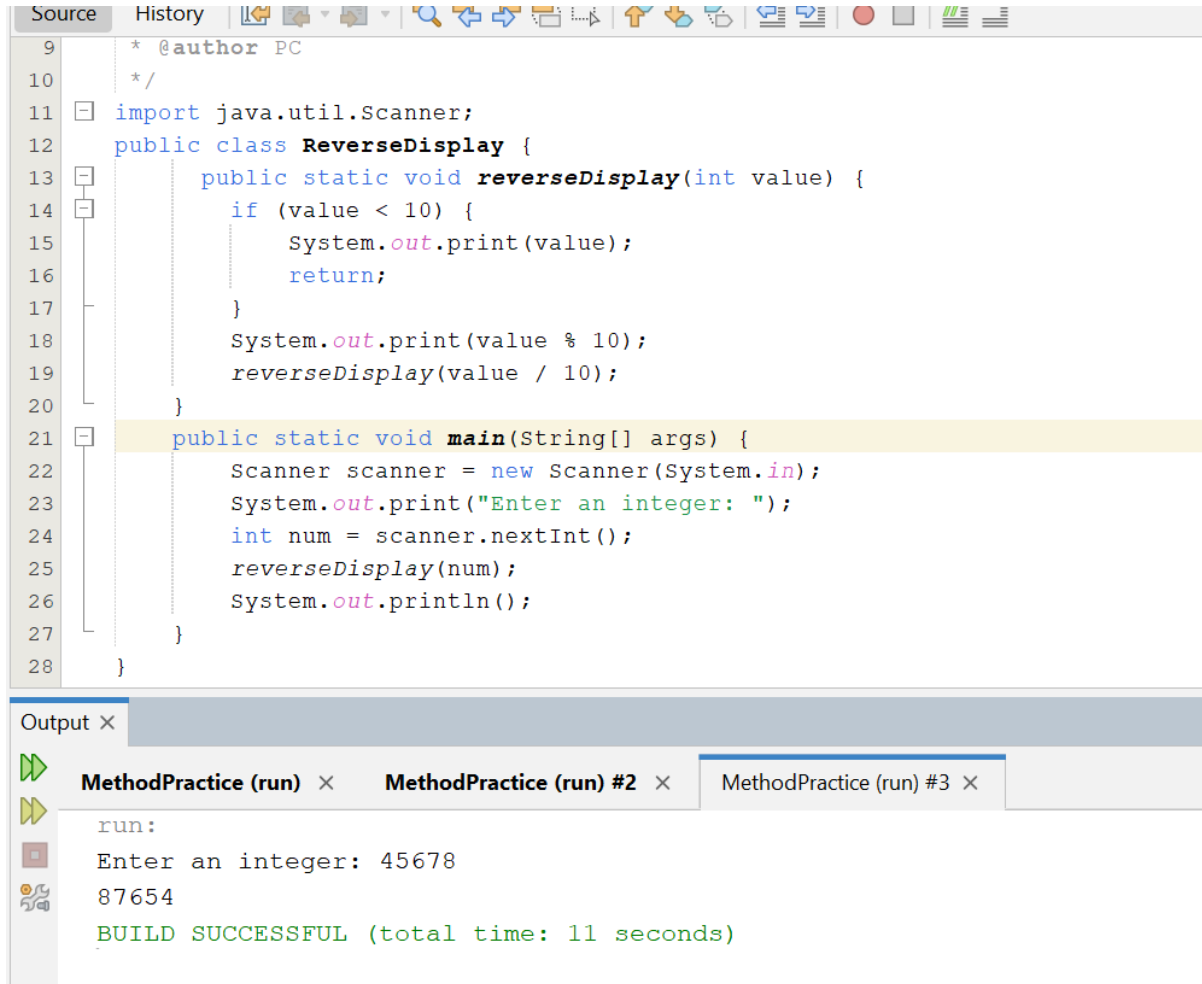
enter a index for a fibnocci series

3

the fibonacci number at index3is2

BUILD SUCCESSFUL (total time: 8 seconds)

Question No 19



The screenshot shows an IDE with a Java source file. The code defines a `ReverseDisplay` class with a recursive `reverseDisplay` method and a `main` method. The `main` method uses a `Scanner` to take an integer input and prints the reversed number. The output window shows the execution results for three runs.

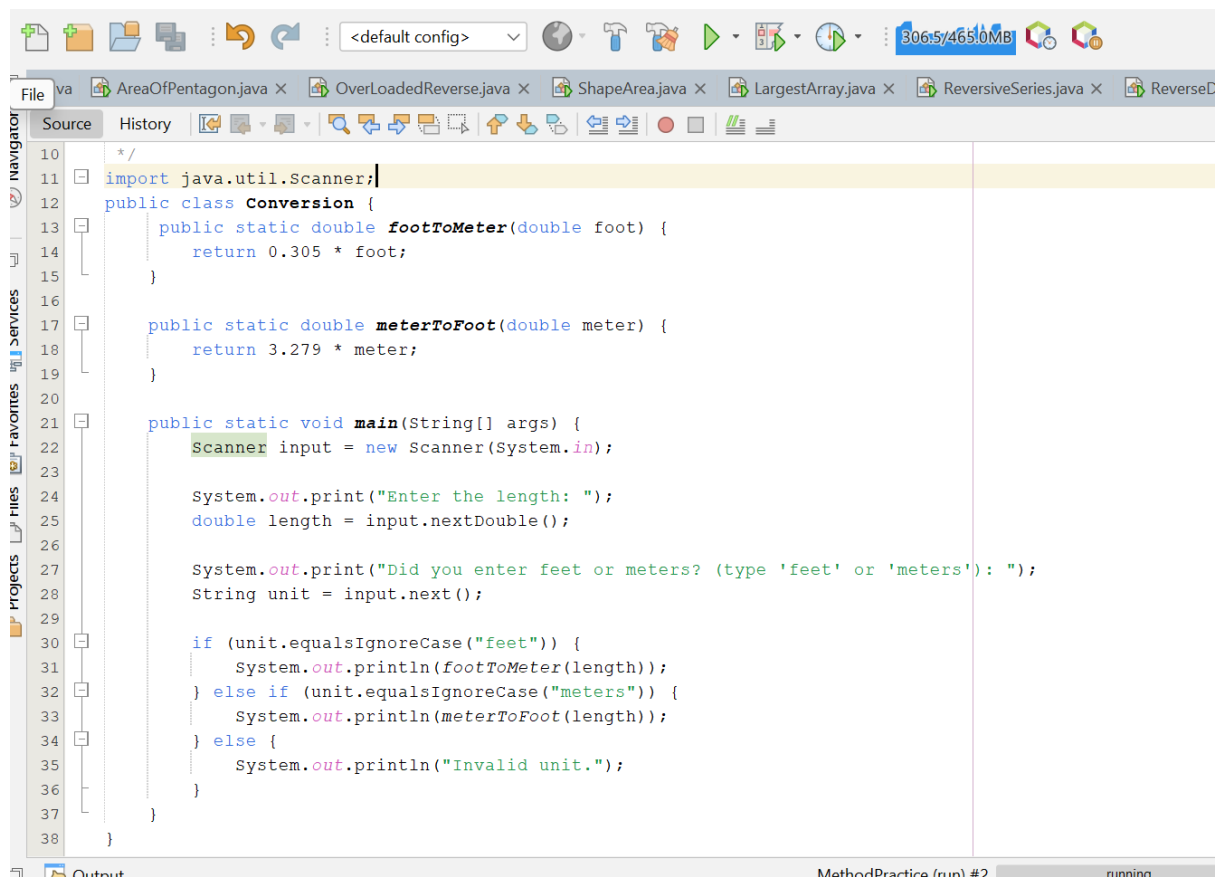
```
9  * @author PC
10 */
11 import java.util.Scanner;
12 public class ReverseDisplay {
13     public static void reverseDisplay(int value) {
14         if (value < 10) {
15             System.out.print(value);
16             return;
17         }
18         System.out.print(value % 10);
19         reverseDisplay(value / 10);
20     }
21     public static void main(String[] args) {
22         Scanner scanner = new Scanner(System.in);
23         System.out.print("Enter an integer: ");
24         int num = scanner.nextInt();
25         reverseDisplay(num);
26         System.out.println();
27     }
28 }
```

Output ×

MethodPractice (run) × MethodPractice (run) #2 × MethodPractice (run) #3 ×

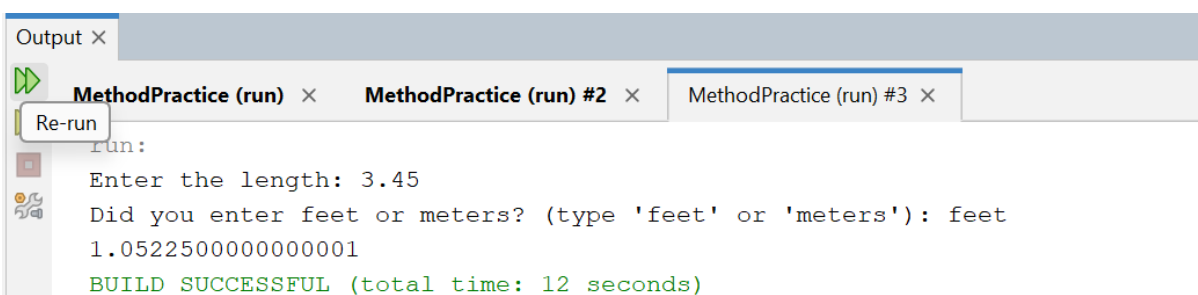
run:
Enter an integer: 45678
87654
BUILD SUCCESSFUL (total time: 11 seconds)

Question No 20



The screenshot shows an IDE window with a Java file named `Conversion.java`. The code defines a `Conversion` class with two static methods: `footToMeter` and `meterToFoot`, and a `main` method. The `main` method uses a `Scanner` to take user input for a length and a unit, then calls the appropriate conversion method and prints the result. The IDE interface includes a toolbar at the top, a file explorer on the left, and a run button at the bottom right.

```
10  */
11  import java.util.Scanner;
12  public class Conversion {
13      public static double footToMeter(double foot) {
14          return 0.305 * foot;
15      }
16
17      public static double meterToFoot(double meter) {
18          return 3.279 * meter;
19      }
20
21      public static void main(String[] args) {
22          Scanner input = new Scanner(System.in);
23
24          System.out.print("Enter the length: ");
25          double length = input.nextDouble();
26
27          System.out.print("Did you enter feet or meters? (type 'feet' or 'meters'): ");
28          String unit = input.next();
29
30          if (unit.equalsIgnoreCase("feet")) {
31              System.out.println(footToMeter(length));
32          } else if (unit.equalsIgnoreCase("meters")) {
33              System.out.println(meterToFoot(length));
34          } else {
35              System.out.println("Invalid unit.");
36          }
37      }
38  }
```



The screenshot shows the IDE's output window with the following text:

```
MethodPractice (run) #2
run:
Enter the length: 3.45
Did you enter feet or meters? (type 'feet' or 'meters'): feet
1.0522500000000001
BUILD SUCCESSFUL (total time: 12 seconds)
```